

```

import pandas as pd

df = pd.read_csv("clv_data.csv")

print("First 5 Rows:\n")
print(df.head())

print("\nDataset Information:\n")
print(df.info())

print("\nShape:", df.shape)

print("\nMissing Values:\n")
print(df.isnull().sum())

df = df.drop_duplicates()

numerical_columns = df.select_dtypes(include=['int64',
'float64']).columns

for col in numerical_columns:
    df[col] = df[col].fillna(df[col].median())

print("\nMissing Values After Cleaning:\n")
print(df.isnull().sum())

print("\nData Types:\n")
print(df.dtypes)

print("\nStatistical Summary:\n")
print(df.describe())

df.to_csv("cleaned_clv_data.csv", index=False)

print("\nPreprocessing Completed Successfully!")

```

First 5 Rows:

	Unnamed: 0	id	age	gender	income	days_on_platform	city \
0	0	0	NaN	Male	126895	14.0	San Francisco
1	1	1	NaN	Male	161474	14.0	Tokyo
2	2	2	24.0	Male	104723	34.0	London
3	3	3	29.0	Male	43791	28.0	London
4	4	4	18.0	Female	132181	26.0	London

	purchases
0	0
1	0
2	1
3	2
4	2

Dataset Information:

```
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            5000 non-null   int64
1   id                    5000 non-null   int64
2   age                   2554 non-null   float64
3   gender                5000 non-null   str
4   income                5000 non-null   int64
5   days_on_platform      4859 non-null   float64
6   city                  5000 non-null   str
7   purchases             5000 non-null   int64
dtypes: float64(2), int64(4), str(2)
memory usage: 312.6 KB
None
```

Shape: (5000, 8)

Missing Values:

Unnamed: 0	0
id	0
age	2446
gender	0
income	0
days_on_platform	141
city	0
purchases	0

dtype: int64

Missing Values After Cleaning:

Unnamed: 0	0
id	0
age	0
gender	0
income	0
days_on_platform	0
city	0
purchases	0

dtype: int64

Data Types:

Unnamed: 0 int64
id int64
age float64
gender str
income int64
days_on_platform float64
city str
purchases int64
dtype: object

Statistical Summary:

	Unnamed: 0	id	age	income
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	2499.500000	2499.500000	30.103200	79593.336200
std	1443.520003	1443.520003	8.668701	60304.056095
min	0.000000	0.000000	10.000000	4.000000
25%	1249.750000	1249.750000	30.000000	32752.500000
50%	2499.500000	2499.500000	30.000000	65989.000000
75%	3749.250000	3749.250000	31.000000	115406.500000
max	4999.000000	4999.000000	50.000000	388572.000000

	purchases
count	5000.000000
mean	1.101000
std	1.181473
min	0.000000
25%	0.000000
50%	1.000000
75%	2.000000
max	6.000000

Preprocessing Completed Successfully!